

SECTION 13892-A GAS DETECTION SYSTEM (GDS)

PART 1 - GENERAL

1.1 Definitions

- A. Contractor: This term is used for the vendor who wins the bid and will be responsible for this package.
- B. Specialty Vendor: This term is used for the vendor(s) who will be responsible for independent systems such as Bulk Gas, Chemical, WWT or UPW, HVAC, C/R. Their responsibilities include furnishing and installing the PLC and necessities to monitor and control their systems. Contractor belongs to Specialty Vendor.
- C. SCADA system: The control and monitoring system provided by Contractor.
- D. Owner: The Micron authorized persons or engineers who are responsible for this project.
- E. FMCS: This term is used to describe the system in which the Owner will use to operate and monitor the Fab, supporting and other buildings. When all Specialty Vendor Systems are integrated into the FMCS, an operator will be able to access each system from a single monitor.
- F. FMCS Integrator: This term is used for the vendor who will be responsible for installing hubs, servers, operator workstations and the Ethernet backbone in FMCS room. This vendor's responsibility will be integrating each Specialty Vendor System (e.g. Bulk Gas, HVAC) into the factory wide platform.
- G. HMI: It means touch panel in the field.
- H. Operator Workstation (OS): This term is used to describe the physical computer terminal, or display, or operators' TFT LCD Panel in the control room for operator monitoring and control related system.

1.2 Project Description

This specification describes a portion of the GAS DETECTION System (GDS). Specialty vendor or contractor shall also

coordinate with the overall FMCS integrator to understand the network and interfacing for this project.

- A. The GDS get the data of gas detector from tool side (including gas box, pump, scrubber) and environment in Fab area.
- B. The control output would trigger tower light and broadcasting for alert to persons and reduce disasters if gas leak occurred.
- C. The GDS architecture shall be based on distributed equipments with centralized concept manipulation of fully intelligent, stand-alone operating in a multi-tasking, multi-user environment on a true peer-to-peer, TCP/IP Ethernet LAN.
- D. The Control and Monitoring system shall be a PC-based system with common SCADA software. By using this software function, the SCADA can reach all the employees' requirement by PLC and SCADA operation. It can access real time data through SCADA software driver to PLC or controllers. The accessed data will be stored on the SCADA server for serving Viewers.

1.3 Work included

- A. Provide hardware, installation, cabling, connection, and software engineering, designing, setting, database build, graphic generation, report generation, testing, commissioning of defined scope including, but not limited to, the following:
 - 1. Two sets of single PLC.
 - 2. Communication cables from PLC to Edge Switch.
 - 3. Software Engineering (database build, point setting and linkage, dynamic graphic generation, alarm management, history trending, report generation etc.)
 - 4. Gas detectors.
 - 5. Sampling pipe and cap are provided and installed.
 - 6. PTL (tower light and buzzer).
 - 7. Conduit, cable tray, supporting and air tube.
 - 8. Control panels/cubicles and marshalling panels.
 - 9. PDP and power lines.
 - 10. Hardware installation, cabling, wire connection.
 - 11. Testing and commissioning.
 - 12. Training
- B. Planing and designing for sampling points location, power demand and usage management.

- C. Provide Liquid leak system by turn key, include providing and designing of hardware, software and related engineering.
- D. The Contractor shall cooperate with FMCS Integrator in development hardware/ software procedure or method and in function testing to complete the whole system integration. The coordination with GMS package for interfacing, operation tuning and commissioning is also important.
- E. Contractor must participate the FMCS integration meeting and obey the related specification or conclusion from user and FMCS Integrator for system integration.
- F. Software and operating system version are complied with document "FMCS integration guideline" provided by FMCS integrator. The related software and hardware development tools including SCADA license shall be prepared by Contractor.
- G. System backup function, training, operation manual and SOP shall be provided.
- H. Provide a two-year warranty on hardware, software, and firmware for all items provided by Contractor. The warranty period will begin when the system has been completely commissioned.
- I. The calibration of Gas detector and Liquid Detection, spare parts, consumables and necessary maintenance quantity are included within warranty period time.
- J. Calibration of GD, once the warranty period expires, it should perform calibration one year and replace the sensor in the other year, which it will be a two year cycle.
- K. For those GD have PM interference issue, such like CO, TEOS, etc, Under the premise of not to effect the GD function, it should have additional active carbon filter which can strain ethyl alcohol, IPA, 5060 and vaporized liquid.
- L. The door open status, power and inside temperature of control panel shall be monitored.

1.4 Work Not In This Specification

- A. SCADA software

1.5 Codes And Standards

- A. All materials, components, and devices shall comply with applicable National, County, and local codes and standards, and the following:
 - 1. American National Standards Institute (ANSI)
 - 2. International Organization for Standardization (ISO)
 - 3. Electrics Industries Association (EIA)
 - 4. The International Electrotechnique Commission (IEC)
 - 5. Instrument Society of America (ISA)
 - 6. National Electrical Code (NEC)
 - 7. National Electrical Manufacturers Association (NEMA)
 - 8. National Fire Protection Association (NFPA)
 - 9. Underwriter Laboratories (UL)
 - 10. Electrical Taiwan Local Code (電工法規)
 - 11. All manufacturers' recommended procedures.
- B. The instrument and control system numbering system will generally follow the standards of the Instrument Society of American (ISA) S5.1. Instrument and control valve data sheets will be provided in accordance with ISA Standard S20.
- C. GDS need to follow the latest regulation which the time need to within 60 second between gas leak and detect.

1.6 Submittals

- A. All submittal shall be in accordance with "13809-A Submittal of Monitoring and Control System.", and be approved by Owner. Not only documents mentioned in the 13809-A and described below, but also owner extra requested should be submitted.
- B. Design Document
 - 1. Monitoring & control system architecture
 - 2. Control panel and cable tray layout
 - 3. P&ID
 - 4. Screen display and report format
 - 5. Point database definition and configuration
 - 6. Control valve brochure and schedule
 - 7. Gas detector and instrument brochure and specification data sheet
 - 8. Test plans
- C. As Build Drawings
 - 1. Monitoring & control system architecture
 - 2. Control panel and cable tray layout
 - 3. P&ID
 - 4. I/O assignment
 - 5. Database configuration and point setting (including SP,

- H/L limit etc.)
- 6. Cable and wire connection table
- 7. Control valve schedule
- 8. Gas detector and instrument specification data sheet
- 9. Electrical one line diagram and PDP source
- 10. Panel/ cubicle fabrication drawings
- 11. Test reports
- 12. Original equipment supplier's operation manual and maintenance manual
- 13. Program source code, configuration setting, database disk copy
- 14. License certificate
- D. Operation Manual of monitoring and control system
 - 1. Hardware structure
 - 2. System operation procedure
 - 3. Software configuration and setting
 - 4. Screen and report operation
- E. Maintenance Manual of monitoring and control system
 - 1. Hardware
 - 2. Software and database
 - 3. Communication

1.7 Responsibilities

- A. There is no price reapportionment under the turn-key work scope unless the bidding quantity is not matched.
- B. All material, equipment including brand, model no. etc. shall be submitted before purchasing for owner approval.
- C. All design and construction method/procedure document shall be provided before construction commencement for owner approval.
- D. Any material, equipment and construction are not complied with approved submittal shall make a replacement immediately with no bargain.
- E. All the ownership and license of delivered subjects including hardware and software, no matter package or self-built, belong to owner. Owner has the right to modify the design and transfer the usage right.
- F. Site management shall follow regulations and local government code to avoid any accident and damages. Contractor should take the responsibility to recover the damage if cause was from his jobs.

- G. Contractor shall improve any problems occurred before warranty termination.
- H. Contractor shall repair those damages from non man-made fault with no extra paid in the warranty period time. Claim could be requested by contractor if damages were caused from man-made fault. The contractor has the responsibility to repair all the faults in warranty time with no excuse.
- I. Experienced employee shall stay on site for coordination in whole construction time. All the engineers and subcontractors shall get user permits or approval before works. Contractor must change the unqualified persons by owner request.
- J. Provide a two-year warranty on hardware, software, and firmware for all items provided by Contractor. The warranty period will begin when the system has been completely commissioned.

PART 2 - PRODUCTS

Electro-cell: TBD

Chem-cassette: TBD

FTIR: TBD

GD shall be remain 20% of all demand points for future expansion.

2.1 General Requirements

- A. The control architecture of life safety related control loops must follow the conclusion of PHA.
- B. The control loop interlock with equipment in Plant A3 which must be recognized and confirmed by Micron staff, shall be installed on the existing FMCS control system under the instruction of Plant A3 working rules.
- C. For those "package control system", control system had built-in as an independent package, shall be setup or installed on the FMCS under the agreement of user. The package control architecture shall be provided for approval.
- D. Control and operation will not be affected and without loss of function when FMCS is out of order.
- E. The display of OS, HMI and reports will be in Chinese.
- F. All equipments, materials and development software/license shall be user approved.
- G. All system software will be based on Microsoft latest operating system or user approved version.

2.2 Programmable Logic Controller (PLC)

- A. No hardware or software can be tailor-made for this project. Only field-proven commercialized available PLC products will be used, such as Siemens. The detail description is in the specification section 13452-A. Any conflict with this paragraph shall obey the sentences below.
- B. Redundant structure should be considered when "critical control subsystem" are assigned by system owner. Controller will be redundant on CPU, power module and remote I/O communication cable. For those assigned "critical control loops" shall be installed on the redundant PLC architecture but also with redundant I/O modules. The "cancel" in the notes column means redundant I/O module is not necessary.

- C. It could be controlled or operated at local site while PLC or controller was out of order.
- D. The interlocking from different I/O points shall go into same PLC or controller since the communication for interlocking is not allowed.
- E. The I/O signals shall follow the specification below,
 - 1. DI: dry contact at field driven by 24VDC from I/O cabinet
 - 2. DO: relay coil at field driven by 24VDC from I/O cabinet
 - 3. AI: 4 to 20 ma differential type, isolated and individual fused. (except RTD)
 - 4. AO: 4 to 20 ma differential type, isolated and individual fused.
- F. The PLC and controller shall have latch and momentary functions for DO signal. Output relay is necessary for DO control signal.
- G. The warning signal shall be designed and indicated if 4 - 20mA signal communicate error or other mal-functions.

2.3 HMI Local Touch Panel

- A. 10.4" backlit TFT LCD touch screen for HMI operation.
- B. Resolution 800*600 pixels at least.
- C. 256 colors at least.
- D. Communication: Ethernet or RS485 or PLC vendor's fieldbus.
- E. Memory should be enough for application with flash memory.
- F. Protection grade: IP65 or NEMA4 (indoor use)
- G. Fanless cooling system.
- H. Ambient temperature and humidity: 0~50°C, 5~85% non condensing.

2.4 Control Panel/Cubicles and Wires

- A. The detail description is in the common specification section. Any conflict with this paragraph shall obey the sentences below.
- B. The control panel shall equipped dual loop power supply, one is emergency power, and the other is UPS power. UPS power will be used in normal operation.
- C. Cubicles will be NEMA 4X for wet, damp and corrosive

locations; NEMA 12 for all other locations. The cabinet shall have cooling fans for temperature control inside cubicle.

- D. All field wiring shall enter into the cubicles from bottom of cubicle to prevent condensate for wet and damp areas, or with a roof cover to avoid. Top entering is acceptable for dry area which is impossible to drop water.
- E. All field wiring will enter into the boxes from the bottom to prevent condensation in wet and damp areas, or be provided with a roof cover to avoid spills from above.
- F. The digital signal wire dimension shall be at least 1.25mm^2 with shielding and analog signal wire shall be at least 0.75mm^2 with copper shielding (Single pair must be 1.25 mm^2 instead). All wires or cables shall go through conduit or cable tray. The conduit must be EMT pipe at internal room and RSG pipe at outside.
- G. The door open alarm of control panel shall be provided. LED lamp shall be installed and power receptacle will not be allowed inside CPU panel.
- H. The power low alarm message shall send to SCADA system when 20V beneath.

2.5 Gas Detector and Instrument

- A. Gas detector are specified in other document.
- B. Explosion area will use explosion type sensors and transmitters. The installation shall comply with the explosion code.
- C. The 4 to 20 ma analog signals are industrial standard. Two-wire type, capable of driving a load at least 600 ohm, is adopted for easy installation and maintenance. Some devices will require four-wire connections, but they will be kept to a minimum.
- D. All control valve could operate manually on site if PLC or controller failed.
- E. The installation method of instrument and consumables must be maintainable and can not be welded for ease replacement.

2.6 Power Meter

- A. Capable of 3p3w and 3p4w data measurement (A, V, W, WH,

Var, VarH, F, PF etc.)

- B. LCD/ LED display.
- C. Support serial communication and data format can be acceptable by Power SCADA system.
- D. Diagnostic function and alarm output.
- E. Harmonic recording and communication to power SCADA.
- F. EMI and noise immunity capability.
- G. Accuracy: V, A $\rightarrow \pm 0.25\%$ FS; W, Var $\rightarrow \pm 0.5\%$ FS; WH, VarH $\rightarrow \pm 0.5\%$ FS.

2.7 Ethernet Cables

- A. The cable between Edge Switch and PLC should be 100Base-T Cat5/5e or Cat6 FTP (foil screened twisted pair) cable. The optical fiber should be adopted instead of FTP if cable length is over 75m.
- B. The optical fiber should be multimode 50/125um with 4C per cable at least or 9/125 um single-mode, depends on distance.
- C. Cables shall be installed inside the EMT pipe. The both side of the FTP cable shall be shield grounding.

2.8 Accessories

- A. Fuse in control system shall adopt ceramic tube type. When sizing, selecting and coordinating fusing for the FMCS system, the following factors shall be considered:
 - 1. Normal operating current - The current rating of a fuse should typically be derated 25% for operation at 25 °C to avoid nuisance blowing.
 - 2. Application Voltage - The voltage rating of the fuse shall be equal to or greater than the available circuit voltage.
 - 3. Ambient Temperature - Derating the fuse shall be considered if the ambient temperature is greater than 25 °C.
 - 4. Overload current and length of time in which the fuse must open.

5. Maximum available fault current - the interrupting rating of a fuse must meet or exceed the maximum fault current of the circuit.
- B. All fusing shall be coordinated to ensure that an over current condition at the instrumentation level will not affect the power distribution to other instruments, PLC modules or power supplies.

2.9 Liquid Leak Detection System

- A. The system can sense water leak and acid, alkaline and organic liquid leak by sensing cable detection. The system architecture has main controller, graphic station with PC and software, LD modules, connectors, terminators, communication cables and sensing cables.
- B. The location, time and cable impedance shall be report to main controller by RS-485 communication when liquid leak detected.
- C. The leak event shall be recorded and displayed the location by graphic method while leak happened. The sensing cable is broken or cut must be alert operator and recorded to report.
- D. The system can be installed in the explosion area.
- E. The sensing cable leak signal, layout and construction shall complied with A3 existing one.
- F. Provide hand-held portable detection instrument to detect cable broken for leak maintenance.

PART 3 - EXECUTION

3.1 System Capacity and Performance Requirements

- A. The control system will be operated 24 hours a day, 365 days a year. Any maintenance works cannot effect the system operation.
- B. The time between status change in the field and alarm displayed in the Operator Station will be less than 2 seconds.
- C. Analog measurement alarms (exceed limits) will be generated within two seconds of alarm condition occurrence. Normal value updates in the display will be within 3 seconds after occurrence.
- D. Performance or loading shall not be loaded over the following requirement:
 - CPU: 50%
 - Network: 30%
 - I/O max. points of each PLC hardware: 4000 points
 - I/O max. points for each remote I/O station: 1000 points

3.2 Spare Requirements

- A. No spare I/O modules are necessary.
- B. Additional space of control panels and I/O cabinets will be 20% with no extra terminals installed.

3.3 Risk Deduction Requirements

- A. The signal lines from redundant sensors shall connect to different I/O modules on different chassis for risk consideration.
- B. The same physical signals from same group devices shall distribute to different I/O modules to avoid one I/O module failed to influent same group signals while I/O assignment design.
- C. Fuse and NFB shall not be installed between power supply and branch of use.

3.4 Software Engineering

- A. The SCADA application software and all graphic displayed shall be designed according to owner requirement or needs (For example, GD overview display, operation pages, and one-stop isolation function key for isolating GD from gas room.).
- B. The diagnostic status and alarm signals of each server, communication device, controller and I/O shall be provided and displayed on Operator Station (OS).
- C. The PLC and remote I/O shall be monitored for power loss, communication loss and diagnostic capabilities that are capable of alarming to control room.
- D. Alarm management (includes commissioning and operation stage) and abnormal handling shall be designed in control system. The field control function shall not be affected if FMCS and SCADA controllers are out of order.
- E. The naming rule and OS/HMI development (screen, format, color etc.) shall accord to document "FMCS integration guideline" prepared by FMCS Integrator.
- F. Time in PLC shall synchronize with FMCS network or GPS time.
- G. Alarming at FMCS layer is acceptable. Alarms at PLC layer may only be needed if an interlock requires an alarm to be generated.
- H. The diagnostic messages of CPU, power module, communication, and I/O modules shall send to control room for monitoring and alarming. The status of each module and system communication situation shall be monitored and displayed the graphic location on the OS in control room.
- I. Report system will be generation by user requirement. The user can on-line operate the adding, deleting and modifying functions.

3.5 Site Works and Management

A. Site Layout Design

1. The location of panels, cable trays, sensors, control valves etc. shall be approved by owner.
2. All cable route and equipment layout design shall consider about EMI to avoid interference.
3. Route layout shall consider function, manageability and good looking.

B. Piping, Cabling and wiring

1. Construction method shall comply with regulations and local codes.
2. Identifier ring in two ends of cable and wire is necessary. The naming rule and color shall consistent with FMCS integrator required, and shall submit to owner for approval.
3. The color code of power lines, signal lines, control lines and network lines shall be in accordance with owner's regulations.

C. Installation

1. The power source of control devices, sensors, control valves etc. shall adopt two independent sources.
2. Control power shall come from independent power branch and shall not branch to other purpose using.
3. Equipment installation shall follow equipment suppliers' recommended instruction.

D. Grounding

1. The grounding requirement is in the specification section "16060-A Grounding". Any conflict with this paragraph shall obey the sentences below.
2. Grounding line shall adopt multiple strands copper wire with O type connecting terminations.
3. Equipment grounding shall follow equipment suppliers' recommended instruction.

E. Safety and Environment Cleanness

1. All the safety management persons and environment cleanness persons shall be organized as owner's requirement.
2. The execution rules and cost apportioned shall obey site regulations from Micron.

3.6 Testing

- A. The test plans shall be submitted with the required 14 days prior to the scheduled testing for owner approval. The plans shall define all the tests required to ensure that the system meets technical, operational, and performance specifications.
- B. The Contractor is responsible for providing all personnel, monitors, equipment, instrumentation, and supplies necessary to perform all testing.
- C. The test plans shall include, not limited to, target, location, schedule time, resources such as equipment, personnel, facilities, and supplies required, the capabilities and functions, test procedures and method/simulation including detailed instructions for test setup, execution, and evaluation of test results, and test report form.
- D. Reports shall be delivered to Owner within 7 days after completion of each test.

3.7 Training

- A. The Contractor shall submit the training plan for owner approval. Approval of the planned training schedule shall be obtained from the Owner at least 30 days prior to the training.
- B. The plans shall include, not limited to, target, courses, location, schedule time, trainee number of each course, facility resources etc.
- C. Training course shall include, not limited to, general hardware architecture, engineer configuration, operation and maintenance.

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